

BASIC SIGHT-SINGING AND EAR TRAINING FOR CHURCH SINGERS

UNIT 8.2 – PITCH BASIC HARMONY – DEGREES OF THE SCALE AND TRIADS



HARMONY: A LITTLE BIT OF HISTORICAL BACKGROUND

The evolution of modern-day Western **harmony**—the type encountered most often in Orthodox choral music—was a process that took many centuries. In the course of much experience and experimentation of working with melodic lines (two, three, four and even more), composers identified certain stable vertical combinations of three or more notes sounding together—referred to as **chords**.

The principles of chord formation and the relationships between them were codified in 1722 by French composer Rameau, in his *Treatise on Harmony*, but by then chords and chordal thinking had been in use for at least a century, perhaps two.

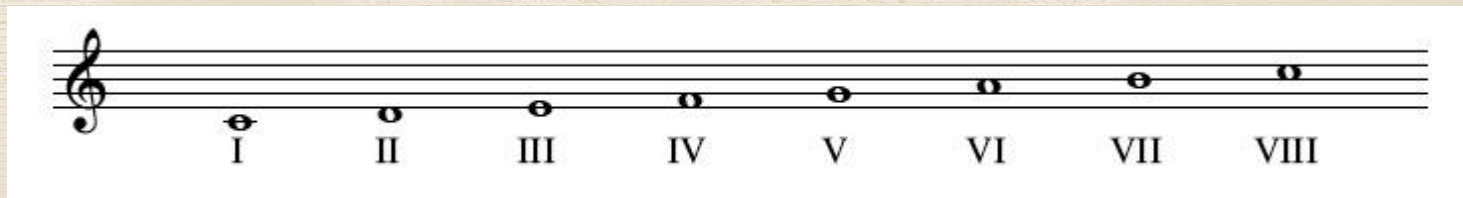
Harmony came into the Orthodox Church from Western Europe, where it had initially developed. The style of singing church hymns in harmony, which has developed in the Roman Catholic Church and among the Protestant reformationists, was adopted in the early 1600s by the Orthodox in Ukraine. Initially, this was a missionizing and pastoral device, to prevent Orthodox believers from being attracted

to the Uniate churches under Roman influence. From Ukraine, harmonic singing came to Moscow in the 1650s, where it was favorably received by Tsar Alexis and Patriarch Nikon, and proceeded to spread rapidly throughout the Russian Church, particularly after the Old Ritualist schism. (Among other things, the Old Ritualists or Old Believers resolutely rejected singing in harmony, instead preserving unison znamenny chant to this day.) In 1668, two Orthodox patriarchs, Paisios of Alexandria, and Makarios of Antioch, who were visiting Moscow, decreed that singing in harmony was in no way unacceptable in Orthodox worship. And the rest... is history, as they say.

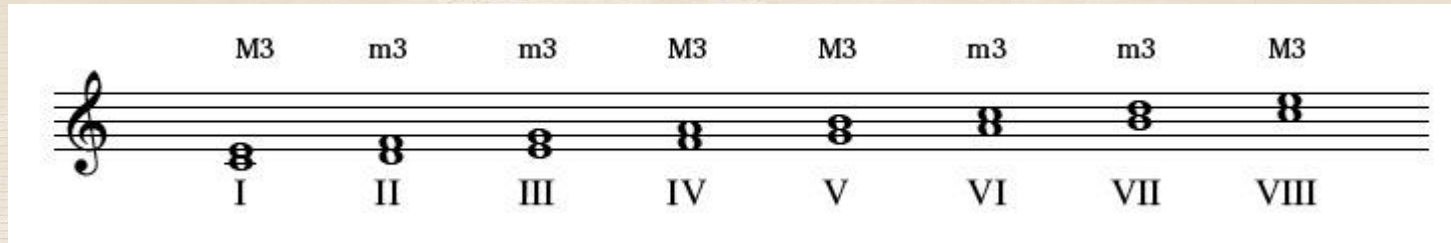
The Russian Orthodox Church went on to develop one of the richest repertoires of harmonic choral singing of any nation, including both harmonic arrangements (harmonizations) of chant melodies and free compositions by a multitude of composers. This repertoire continues to be sung in Russian churches as well as in concert halls and sound recordings, sometimes becoming an important missionary outreach to the Western world.

DEGREES OF THE SCALE

To illustrate the practical aspects of harmony that are most important to choral singers in the Orthodox Church today, we will define—in several stages—some basic concepts, which build upon the knowledge of intervals and scales already covered in this course. The first is the concept of **degrees of the scale**. To indicate these “degrees,” each note of the scale (in this instance, the C major scale) is assigned a Roman numeral, I through VIII. Just as relationships between individual notes is expressed using solfege syllables, the relationships between chords, called **chord progressions**, are expressed by Roman numerals.



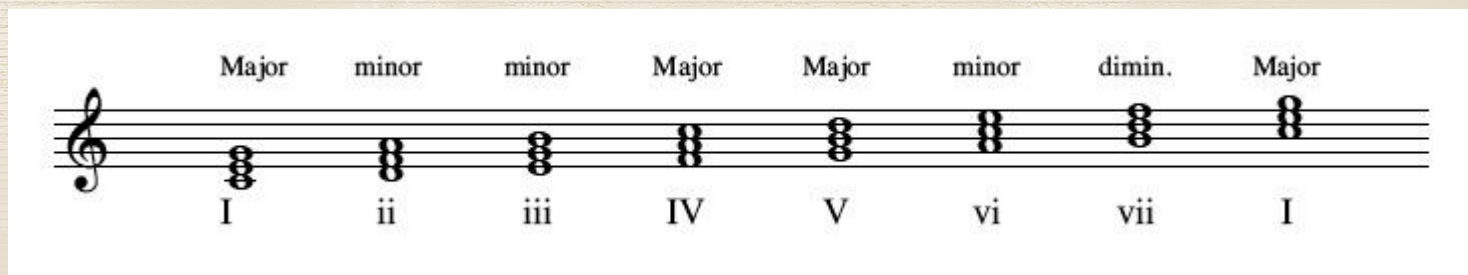
As the next step towards creating harmony, we add another voice, **in parallel thirds**, above the notes of the C major scale:



These two-note combinations are not yet full-fledged chords, but they already possess a very important element—the interval of the third above the bottom note—which will determine the chord’s **quality** or **character**. Some thirds—marked **M3**, with a capital letter—are **major**, while others, marked **m3**, are **minor**.

TRIADS OF THE MAJOR SCALE

Continuing the process of adding notes vertically, we now add another voice, **in parallel fifths** with the bottom notes. This process yields bona fide three-note chords, which, because they consist of three notes stacked vertically, are called **triads**. Depending on what kind of third, major or minor, was already present at the bottom, the resulting chord is either a **major triad**, or a **minor triad**. The exception is the triad on the VIIth degree: because the interval from B to F is a diminished fifth, it is called a **diminished triad**. It is common practice to use **capital** Roman numerals to signify major triads and **small** Roman numerals to signify minor and diminished triads.



SPELLING THE TRIADS

Notice that

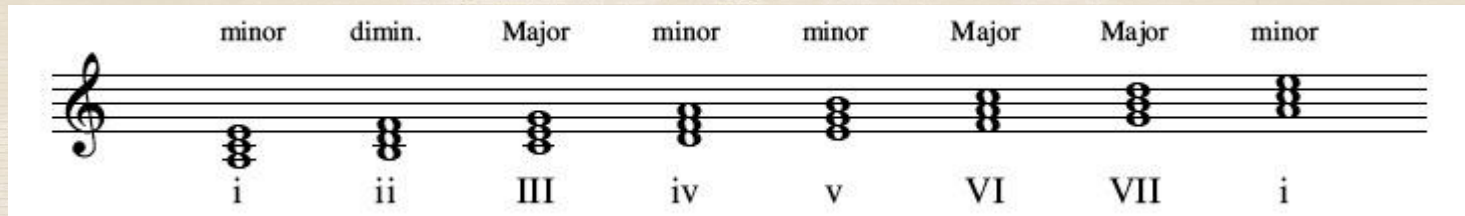
- ◆ a major triad consists of a major third on the bottom and then a minor third
- ◆ a minor triad consists of a minor third on the bottom and then a major third
- ◆ a diminished triad consists of a minor third on the bottom and another minor third

For the choral singer it is extremely useful to learn the “spellings”—in SOL-FA syllables—of the various triads, e.g.,

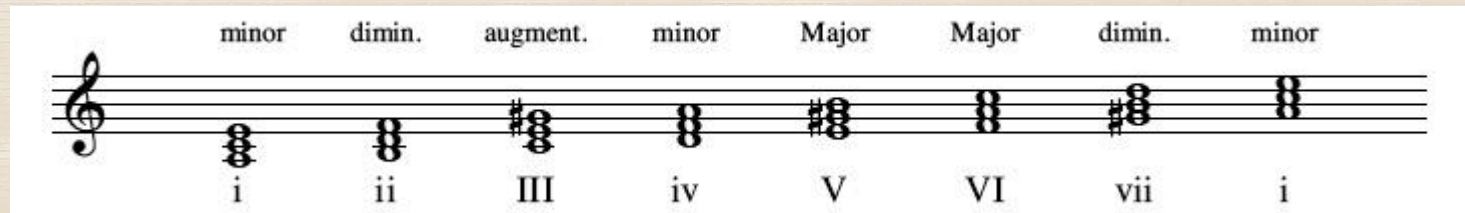
I	=	DO-MI-SOL
ii	=	RE-FA-LA
iii	=	MI-SOL-TI
IV	=	FA-LA-DO
V	=	SOL-TI-RE
vi	=	LA-DO-MI
vii	=	TI-RE-FA

TRIADS OF THE MINOR SCALE

A similar set of triads can be constructed using the notes of the minor scale. If the **A natural minor scale** is used, the following series of triads is obtained:



However, as noted in Unit 6.2 (Slides 6 & 7) because of the way chord relationships developed historically, a requirement emerged that the triad on the **Vth** degree, referred to as the **dominant**, be a major triad. This necessitated the sharpening of the seventh scale degree—making the G a G $\sharp$ —and resulting in the **harmonic minor scale** (see Unit 6.2). The triads constructed using the notes of the harmonic minor scale are as follows:



The character of these triads is somewhat different than for the natural minor:

- ◆ the **III** chord is now an **augmented triad** (because of the augmented fifth—C to G#)
- ◆ the **V** triad is now **major**
- ◆ the **vii** triad is now **diminished**

As with the triads of the major scale, it is very important to learn the spellings of the triads:

i	=	LA-DO-MI
ii ^{dim}	=	TI-RE-FA
III ^{aug}	=	DO-MI-SI
iv	=	RE-FA-LA
V	=	MI-SI-TI
VI	=	FA-LA-DO
vii ^{dim}	=	SI-TI-RE

These same triads can be constructed in all the different major and minor keys, and they will be the same in character—major, minor, diminished, or augmented. Learning to recognize the standard major triads (e.g. DO-MI-SOL, FA-LA-DO; SOL-TI-RE) and minor triads (e.g. LA-DO-MI, RE-FA-LA, MI-SOL-TI) in the most common keys (F major/D minor, G major/E minor, D major/B minor, Bb major/G minor) is one of the most important skill sets a choral singer can develop.

USING SOLFEGE TO DISTINGUISH TRIADS

If you want to sing a **major triad** starting with a certain note, call the first note DO, then think and sing: DO - ↑MI - SOL - ↓MI - DO (up, then down)

If you want to sing a **minor triad** starting on that same note, call the first note LA, then think and sing: LA - ↑DO - MI - ↓DO - LA

You could just as easily call the first note of a major triad FA, then sing FA-LA-DO, or call it SOL, then sing SOL-TI-RE. Or, for a minor triad: call the first note RE, then sing RE-FA-LA, or call it MI, then sing MI-SOL-TI. Practice doing this.

LEARNING TO SING TRIADS IS USEFUL FOR:

- ◆ Giving the starting pitch to the choir
- ◆ Establishing the tonality (major or minor) of a piece before you sing it
- ◆ Finding the correct interval when the melody involves leaps of a third or more:
 - ◆ To find a minor third, start singing a minor triad
 - ◆ To find a major third, start singing a major triad
 - ◆ To find a fourth, start singing a major triad and add a half step
 - ◆ To find a fifth, sing a triad and leave out the middle note
 - ◆ To find a sixth, sing a triad and add a half step (for a minor sixth) or a whole step (for a major sixth)
 - ◆ To find a seventh, sing a triad and add a minor third (for a minor seventh) or a major third (for a major seventh)

Please proceed to **PITCH EXERCISES 8.2**

Ut Queant Laxis (Hymn to St. John the Baptist)

Guido of Arezzo
(circa 991-1033)



Ut que - ant la - xis, Re - so - na - re fi - bris, Mi - ra
ges - to - rum, Fa - mu - li tu - o - rum, Sol - ve pol -
lu - ti, La - bi - i re - a - tum, Sanc - te Jo - han - nes.

Translation:

So that your servants may, with loosened voices, resound the wonders
of your deeds, clean the guilt from our stained lips, O Saint John.